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The *Enterarchon*: An ancient visceral brain

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The gastrointestinal (GI) tract is innervated by intrinsic neurites and extrinsic pathways linking it bidirectionally to the central nervous system. These circuits regulate digestion, metabolism, and homeostasis across metazoans. Comparative evidence from early-diverging animals to *Drosophila* and vertebrates suggests gut-embedded circuits arose on an epithelial/peptidergic scaffold and evolved in parallel across lineages, coordinating motility and secretion before centralised brains emerged. While vertebrates offer detailed molecular atlases of enteric nervous system (ENS) cell types, flies allow linking molecular identity with physiological function at cellular resolution. We highlight how the intestine-ENS axis integrates endocrine and immune signals and undergoes remodelling along three axes: sex as a constitutive dimorphism, reproduction as a reversible plasticity, and ageing as a biphasic trajectory. This comparative view challenges brain-centric models of systemic regulation. We propose instead that the intestine, through its neuronal, epithelial, and immune components, acts as an *Enterarchon*: an ancient visceral brain shaping organismal homeostasis.

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Enterarchon, an instrument for survival

Transforming environmental nutrients into internally useable energy is a universal feature of life. In early metazoans, epithelial invagination produced a primitive internal tube—the ancestral *Enterarchon* (BOX1)—whose specialised epithelium established a barrier while allowing absorption, creating a protected internal milieu for nutrient uptake (Figure 1) [1]. This internalised yet externally exposed milieu later diversified into the gastrointestinal (GI) tract—*the enteron*—a structure central to digestion, defence, homeostasis, and reproduction [2,3].

From the simplest metazoans to complex vertebrates, this survival-critical role has made the gut a major site of neuronal innovation [4,5]. How nervous systems evolved to regulate this interface clarifies why the enteric nervous system (ENS)—an ancient, semi-autonomous network—should be considered alongside central neuronal and immune systems. Its interactions with these systems, and the consequences of its dysfunction, highlight the need to decentralise the brain's traditional dominance in models of systemic regulation.

Enteric evolution: ancient scaffold, convergent solutions

Could the ENS be the "first brain"?

The ENS is an ancient, semi-autonomous division of the nervous system that regulates intestinal function across metazoans. ENS-like nets are present in early-diverging animals such as cnidarians (e.g. *Hydra*), where gut-associated circuits mediate peristalsis, expulsion, and feeding responses [6]. These ancestral networks exhibit conserved neuronal features, including synaptic transmission and coordination of epithelial-motile programmes, supporting the view that gut-embedded control was among the earliest manifestations of nervous systems in animal evolution [6].

Nevertheless, recent work in *Hydra* reveals both pre-enteric and central-like neuronal populations whose interplay shapes satiety-dependent behaviour [7], consistent with gut-associated circuits forming an early scaffold for nervous systems, with centralising modules arising alongside or in parallel. Thus, the "first brain" framing [6] should be read as a conceptual metaphor: enteric-type circuitry represents a primordial scaffold for neuronal control, with early co-emergence of

BOX 1.

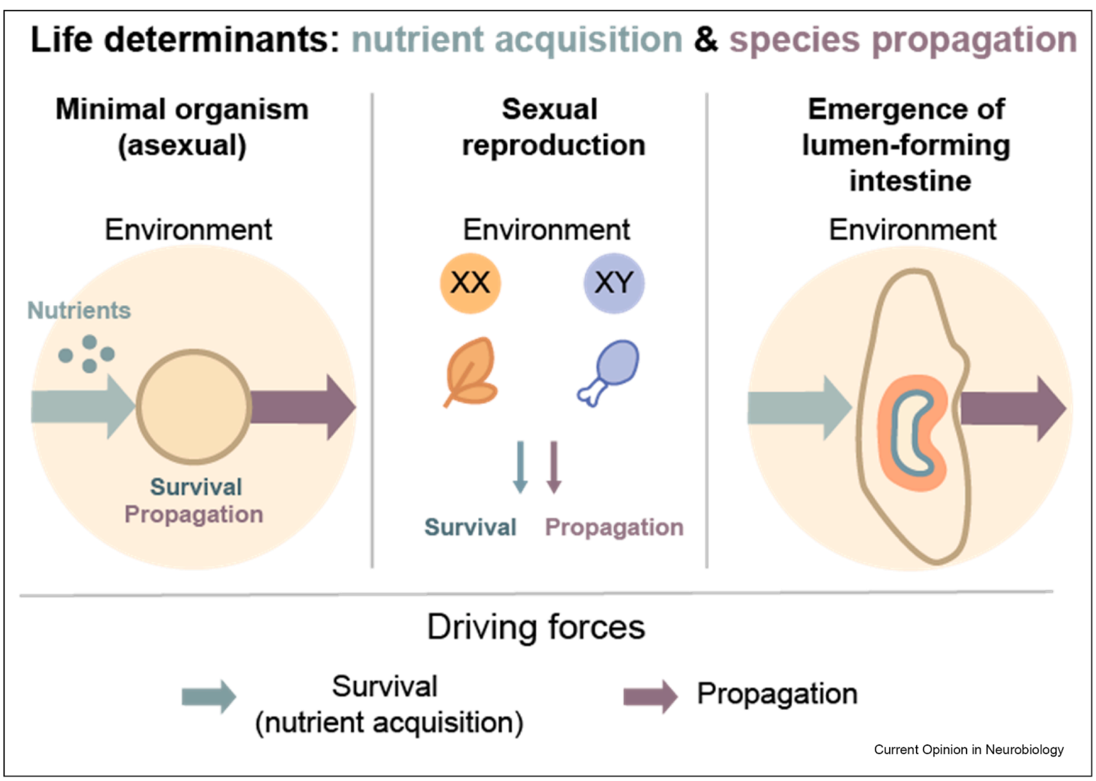
We propose the term *Enterarchon* (from Greek *énteron*, "gut," and *arché*, "origin" or "power") to denote the ancestral intestinal tube that emerged via epithelial invagination in early metazoans. This internalised surface enabled nutrient absorption, immune defence, and homeostatic regulation, laying the foundation for the evolution of complex physiological systems. Beyond its role as "origin", the double meaning of *arché* also highlights the gut as a major locus of physiological control, an *Enterarchon* that integrates energy, immunity, and reproduction from the very beginning of animal evolution. We distinguish this phylogenetic concept from the ontogenetic concept of *Archenteron*, an early intestine emerging during metazoan development.

centralising modules. Comparative evidence for multiple evolutionary trajectories of nervous systems [5,8] highlights that centralised brains were not a prerequisite for neuronal complexity (Figure 2).

Peristaltic responses to nutrients did not always require neurons. In basal metazoans such as placozoans and ctenophores, rhythmic contractions of the digestive cavity are coordinated by ciliated epithelial cells and paracrine peptidergic signalling, suggesting that control of digestion emerged prior to the evolution of synapses and neurons [5]. These epithelial programmes extend beyond digestion to functions such as respiration and immune defence [9]. The ancestral role of the ENS may therefore have been to elaborate and accelerate pre-existing cilia-based motility and secretory circuits.

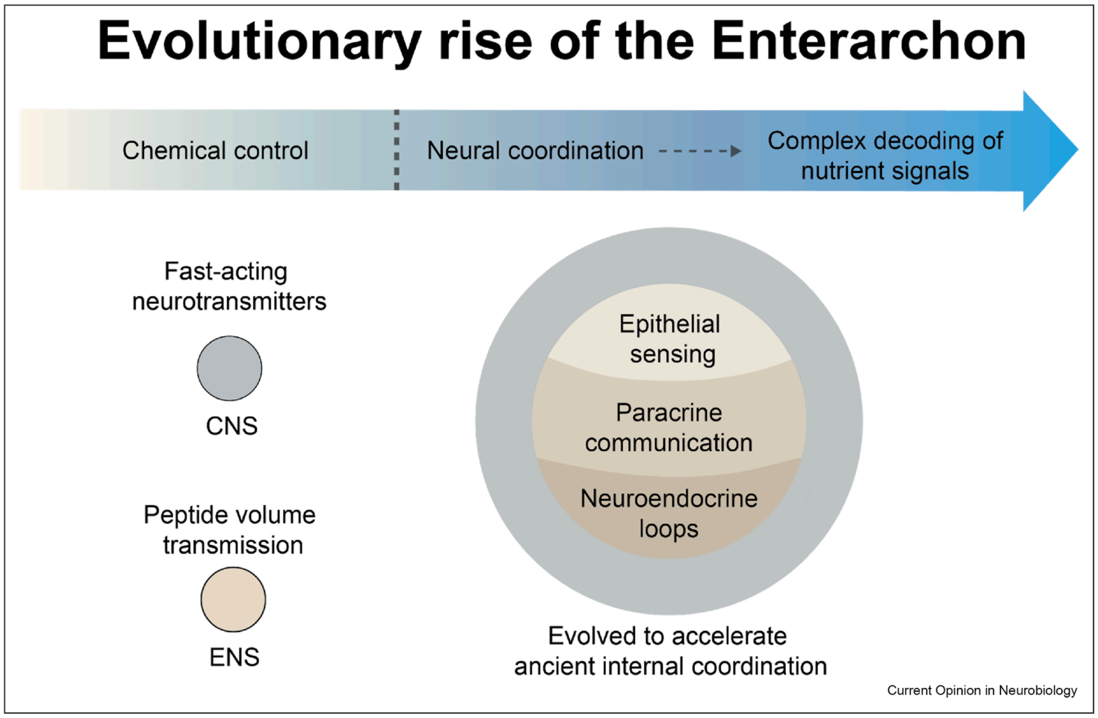
Comparative genomics and phylogeny suggest that neurons regulating internal physiology (ENS-like) may have evolved independently of those mediating external sensorimotor control (CNS-like), with chemical signalling—including peptides and small molecules (glutamate, ATP, and NO)—preceding fast synaptic transmission [4,5]. Such paracrine "volume transmission" likely

Figure 1



Life determinants: nutrient acquisition & species propagation.
Conceptual overview of the major drivers for organismal survival. Nutrient acquisition and propagation shape organismal development, culminating in the *Enterarchon*.
(**a**) Minimal theoretical organism (asexual). A simple unit without a gut or brain is driven by two forces: survival, depicted by the lateral arrow (green) indicating exchange with the environment for nutrient uptake, and propagation. No internal lumen exists.
(**b**) Two-sex system. A minimal theoretical sexual organism experiences the same two forces but partitions strategy by sex (illustrated by two coloured units and an XX/XY cue). Colours denote sex-biased ingestion/digestion strategies (e.g. different food choices or metabolic priorities) under identical ecological cues.
(**c**) Theoretical emergence of a lumen-forming intestine: the *Enterarchon*. To maximise nutrient capture while maintaining an internal milieu, a lumen evolves, creating an inside–outside interface with increased absorptive surface. This epithelial tube becomes progressively regulated and integrated, giving rise to the *Enterarchon*, an ancestral visceral control system that coordinates nutrient uptake and, later, reproductive demands. Green arrows: survival (nutrient acquisition); purple arrows: propagation.

Figure 2



Evolutionary rise of the *Enterarchon*.
Top: A colour gradient arrow summarises the shift from primarily chemical/paracrine control to neural coordination and, ultimately, complex decoding of nutrient signals; the dashed line marks the emergence of coordinated neuronal circuits.
Bottom-left: Communication modes contrasted, with the ENS dominated by neuromodulatory/peptidergic "volume transmission," versus CNS dependence on fast-acting synaptic neurotransmission.
Bottom-right: Schema of conserved gut control layers, including epithelial sensing, paracrine communication, and neuroendocrine/neuronal feedback loops, emphasising that neuronal systems evolved to accelerate ancient internal coordination at the gut interface. Together, these elements support the hypothesis that the GI tract functions as an *Enterarchon*, an ancient visceral brain whose gut-embedded circuits preceded and scaffolded later nervous-system centralisation.

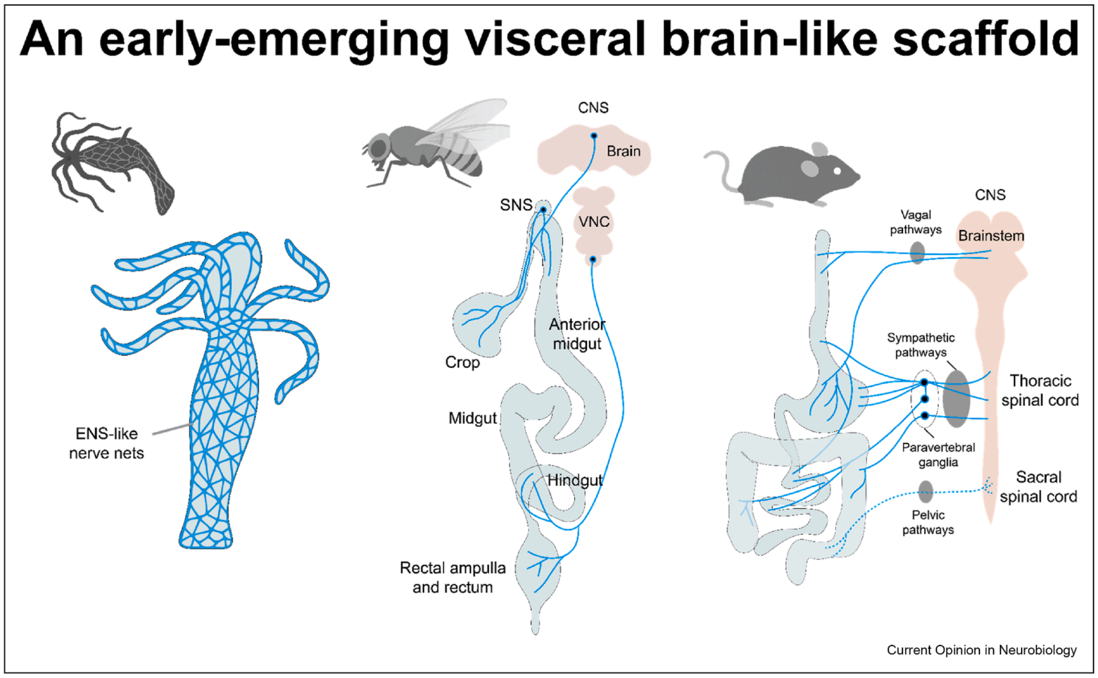
represents an earlier mode of neuronal communication, consistent with the *chemical brain hypothesis*, which postulates that early neurons regulated internal physiology via slow, hormone-like signals before the emergence of fast-acting sensorimotor neurotransmission [2,8]. ENS-like architectures thus likely represent an early manifestation of neuronal integration (Figure 2). Neuronal centralisation—and the emergence of complex brains—appear to have arisen independently more than twenty times across animal phyla [4], supporting a model in which the *Enterarchon* may have provided an early locus of neuronal innovation, with centralising modules evolving later to coordinate distributed functions. Rather than a mere "second brain," the ENS may be an early, brain-like solution for internal-state control—what we term a *visceral brain* (Figure 3), BOX2.

How does the ENS develop across species?

In vertebrates, the ENS derives from neural crest cells—multipotent progenitors that migrate rostrocaudally and colonise the gut [10–16]. Invertebrates lack a neural crest, and ENS-like innervation arises from distinct embryonic origins [17]. In *Drosophila*, three main sources contribute to gut innervation: i) local neuroectoderm neuroblasts in the embryonic ventral nerve cord (VNC), whose axons project to the hindgut. ii) a single dorsal foregut placode that invaginates, subdivides into the frontal, hypocerebral, paraesophageal, and ventricular ganglia of the somatogastric nervous system (SNS) [18], then repositions and fuses during metamorphosis to form the adult SNS; and iii) head neuroectodermal placodes that invaginate late in embryogenesis, fuse with the brain primordium, and generate neuroendocrine neurons of the *pars intercerebralis*.

Thus, whereas vertebrate ENS development depends on long-range neural crest migration and intramural colonisation [19,20], invertebrate ENS-like systems are composite and externalised, with gut innervation arising from multiple embryonic origins [17,18]. Notably,

Figure 3



An early-emerging "visceral brain" scaffold: comparative enteric nervous system wiring across animals.
(Left) Cnidarian (*Hydra*). Diffuse epithelial nerve nets span the body column and tentacles without centralisation, coordinating feeding movements and peristaltic-like contractions through locally coupled neuronal/epithelial activity.
(Middle) Insect (*Drosophila*). The central nervous system (CNS; brain and ventral nerve cord, VNC) communicates with the gut alongside the stomatogastric nervous system (SNS). Extrinsic fibres and intrinsic enteric neurons innervate the crop, anterior midgut, hindgut, and rectal ampulla to control ingestion, storage, transit, and defecation, predominantly under peptidergic neuromodulation. More specifically, the adult fly gut is innervated by three main input sources: i) the SNS, ii) neurons of the CNS located in two discrete regions, namely the descending neurosecretory axons from the *pars intercerebralis* (PI), and projections from the VNC, and iii) neurites of secretory cells of the *corpora cardiaca* (CC) that bundle together with axons of the hypocerebral ganglion (HCG), the largest ganglion of the SNS, via the recurrent nerve (RN), also referred to as the insect vagus nerve. In addition, some local sensory units innervate the rectal ampulla. Together, these inputs form a composite, externalised architecture functionally analogous to vertebrate myenteric and submucosal plexuses.
(Right) Mammal (*Mus musculus*). An intrinsic enteric circuitry in the gut wall interfaces bidirectionally with the CNS through three major extrinsic routes: vagal pathways to the brainstem (parasympathetic), sympathetic pathways from the thoracic spinal cord via paravertebral ganglia, and pelvic pathways from the sacral spinal cord (parasympathetic). These pathways regulate motility, secretion, blood flow, and visceral physiology.
Schematics are not to scale and are simplified to emphasise the conserved motif of an intrinsic gut network coupled with distinct long-range innervation. Abbreviations: CNS, central nervous system; VNC, ventral nerve cord; SNS, stomatogastric nervous system. Figure b and c are adapted from Ref. [70].

BOX 2.

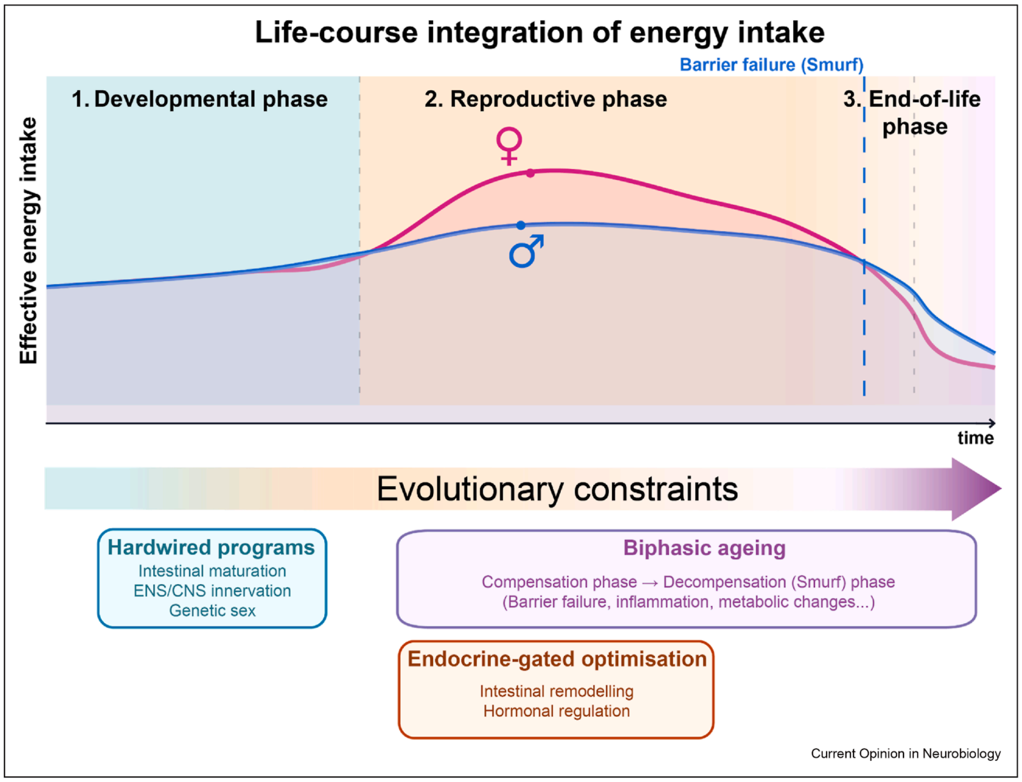
Viscera (Latin, "internal organs") traditionally refers to organs of the abdominal cavity. By describing the concept of a "***Visceral Brain***", we aim to reposition this term beyond its historical association with limbic and emotional processing [69], to emphasise a more ancient neural system: the ENS. Unlike limbic circuits, the ENS autonomously regulates digestion, metabolism, and homeostasis. Its evolutionary origin, functional autonomy, and role in nutrient sensing and inter-organ signalling position it as the most ancient manifestation of a visceral brain.

unlike vertebrates—where enteric neurons are embedded within the gut wall [3]—insects house enteric somata externally, along nerve strands and ganglia adjacent to the gut [21–23], an anatomical configuration that nonetheless preserves core ENS functions. These convergent architectures likely reflect shared evolutionary pressures to ensure rapid, decentralised regulation of digestion and energy balance in response to internal cues.

Did the ENS evolve as an immune-sensing interface?

Beyond coordinating digestion, the ENS may have co-evolved as a neuronal interface for host–microbe communication. Enteric neurons/enteroendocrine pathways can detect microbial metabolites and by-products, providing a direct route by which microbes modulate gut neuronal activity [1,24,25]. Across key developmental windows, the ENS and microbiota co-mature and

Figure 4



Life-course control of nutrient acquisition.
Effective energy intake is plotted across three life stages. Coloured background bands mark [1] the developmental phase of the organism, when hardwired programmes establish gut maturation, ENS/CNS innervation and establishment of sex identity [2]; the phase of reproductive competence, when endocrine cues optimise intake and digestive performance; matching it with reproductive demands [3] Ageing, shown as a biphasic: an early compensatory phase followed by decompensation or the "Smurf state." The magenta curve (female) and blue curve (male) show sex-divergent optimisation during the given reproductive window (female peak reflecting egg-production demands, whereas males show a smaller rise), followed by progressive convergence in late life. The blue vertical dashed line in Stage 2 highlights the "Smurf" phenotype, characterised by a clear drop in energy intake and the onset of intestinal barrier failure, after which the organ function rapidly declines, leading to impending death. Fine dashed lines indicate stage transitions, whereas the horizontal arrow indicates chronologically advancing time. Boxes highlight dominant mechanisms per stage (hardwired programmes; endocrine-gated optimisation; biphasic ageing/Smurf). Curves are qualitative and intended to convey relative trends rather than measurements.

influence one another's structure and function [26]. In early-diverging animals lacking complex immune systems, nascent enteric circuits may have evolved to detect microbial metabolites, virulence factors, or nutrients, offering fast, localised responses at the epithelial interface [5]. These co-developmental dynamics position the ENS as an immunomodulatory, chemosensory buffer tuned to microbial signals, explaining why perturbing early-life microbiota alters ENS maturation and gut physiology [24–26]. Leveraging *Drosophila*, where enteric circuits can be mapped and manipulated, we next highlight conserved principles of enteric control.

The ENS of *Drosophila*: a model for mapping the visceral brain

Invertebrates such as *Drosophila* retain features of visceral autonomy that reflect *Enterarchon*-like organisation. The *Drosophila* ENS is a powerful model for studying visceral control: defined neuronal populations [21,22,27–30], exceptional genetic accessibility [22,31,32], and expanding connectomic resources [33] provide an unprecedented window into the neuronal regulation of gut physiology.

What neuronal types innervate the fly intestine?

The fly GI tract is innervated by central and peripheral neurons (Figure 3). These target discrete compartments of the gut—the ectoderm-derived foregut and hindgut, and the endoderm-derived midgut—via three primary sources: i) the SNS; ii) CNS-derived projections, including descending neurosecretory axons from the *pars intercerebralis* (PI) and projections from the VNC; and iii) processes of *corpora cardiaca* (CC) cells projecting from the HCG along the recurrent nerve (RN; the "insect vagus") to the foregut. Local sensory units also innervate specific sites such as the ampulla. Together, these inputs form a composite, externalised architecture functionally analogous to vertebrate myenteric and submucosal plexuses, enabling region- and state-specific regulation of intestinal function.

What does the Drosophila ENS control?

The *Drosophila* ENS orchestrates feeding, satiety, regeneration, and ionic balance through anticipatory, context-dependent control, integrating sensory cues with internal state to prepare physiological responses [22,27,28,32,34].

Reproduction. Myosuppressin (Ms)-producing enteric neurons innervate the crop and anterior midgut, to regulate gut expandability and food intake after insemination, forming a sex-specific circuit that supports reproduction [22]. Steroidal and gut-derived signals (e.g., ecdysone, Bursicon-a) activate Ms neurons, promoting Ms peptide release onto intestinal muscles. This enhances gut expandability and food intake, and disruption of this loop impairs food intake and fecundity, establishing Ms neurons as integrators of reproductive signals.

Satiety. Enteric $Dh44^{+}$ neurons innervating the crop and anterior midgut integrate nutrient value and satiety signals [27]. Through GLUT1, they distinguish caloric sugars from non-nutritive sweeteners and, when activated, drive ingestion and gut motility. In the fed state, two peripheral satiety pathways inhibit Dh44 activity [35]: i) $Piezo^{+}$ Dh44 neurons that sense crop distension and dampen sugar-evoked responses; and ii) $Hugin^{+}$ VNC neurons that detect internal sugar and release PK2 to act on PK2R in Dh44 neurons, suppressing their activity. Thus, Dh44 neurons integrate internal sugar availability, gut mechanosensation, and neuropeptidergic feedback to regulate feeding.

Ingestion. A gut-brain-gut interoceptive circuit gates sugar intake by nutrient quality [31]. $Gr43a^{+}$ enteric neurons in the HCG detect luminal sugar in the foregut and excite ingestion-promoting IN1s neurons in the SEZ. IN1s then inhibit crop-innervating motor neurons, allowing larger sugar intake. This circuit dynamically couples sugar concentration to internal metabolic state, paralleling vagal gut-brain pathways in mammals.

Anticipatory modulation. Two classes of 5-HT neurons transform taste into anticipatory physiological responses [36]: sugar-responsive SELs activate insulin-producing cells and limit feeding, whereas bitter-responsive SELs project to the HCG, activate $5\text{-}HT7^{+}$ enteric neurons, and drive crop contractions, likely mobilising stored nutrients in anticipation of scarcity. These circuits coordinate motor and hormonal outputs to align digestion with predicted demand.

Regeneration and homeostasis. ARCEN cholinergic enteric neurons sense TNF/Eiger and trigger acetylcholine-dependent epithelial Ca^{2+} waves to restore barrier integrity and resolve inflammation, echoing mammalian cholinergic anti-inflammatory pathways [32,37,38]. Under ionic challenge, INSO gut-projecting neurons detect luminal Na^{+} and induce compensatory salt appetite via gut-to-brain signalling [28], highlighting the autonomy of the ENS in initiating homeostatic responses.

Sex differences in the *Enterarchon*

In sexually reproducing species, males and females adopt distinct physiological strategies to optimise reproductive success. These strategies are encoded hormonally, genetically, and metabolically. In *Drosophila,* the gut is a sexually plastic organ whose geometry, regenerative dynamics, and metabolic profile adapt to reproductive cues [22,32,39]. These programmes are coordinated with ENS input, positioning enteric circuits as regulators of sex-specific physiology.

Why are female guts bigger and more looped?

Female Drosophila exhibit larger and longer guts than males, a dimorphism that becomes more pronounced upon mating [22,40–42]. Internal organ geometry is actively maintained: female guts are longer and more curved than male guts, due to sex-biased expression of bnl/FGF in gut muscles, which sculpts sex-specific tracheal branching and physically imposes gut loops [39]. This reveals inter-organ geometry as an axis of sexual dimorphism with functional relevance. Similarly, in mammals, reproductive remodelling of the maternal intestine is anticipatory and genetically controlled, involving a reversible expansion of the villi, epithelial projections that increase absorptive surface area, and partially irreversible elongation of the intestine, independent of food intake or microbiota changes [43].

Why do female guts regenerate faster?

Male and female midguts contain similar ISC numbers, but female ISCs divide more frequently, sustaining higher turnover and larger guts—particularly during reproduction [40,44]. These effects are consistent with principles whereby local and systemic cues tune ISC proliferation [45,46]. Consistently, disrupting enterocyte adhesion or survival reduces female gut size, indicating greater dependence on continuous renewal [47]. Sex-biased renewal reflects intrinsic and extrinsic drivers: intrinsically, ISC sexual identity—set by a non-canonical branch of the sex-determination pathway—increases proliferative capacity [40]; extrinsically, mating raises steroid and juvenile hormones that acutely stimulate ISC activity [42,48]. Within hours of copulation, ISC divisions increase, enterocyte number rises, and the gut expands, accompanied by transcriptional shifts in lipid handling and nutrient absorption that sustain reproductive demand.

The muscle layer is also sexually dimorphic and steroid-responsive; female muscles remodel to sustain reproduction, a feature conserved in pregnant/lactating mice [41]. Mated females increase intake but defecate less [21], consistent with altered transit/absorption. Therefore, the female gut is a dynamic, hormonally responsive organ that couples nutrient processing to reproduction via a tightly regulated proliferative program.

How is metabolism sex-tuned in the gut?

Transcriptomics of virgin midguts reveals broad sex-specific programmes: females are enriched in programmes related to cell proliferation, whereas males preferentially express carbohydrate/oxidative metabolism genes [41,49]. In females, ISC sexual identity underpins increased proliferation [40]. In males, a testes-to-gut signal activates JAK−STAT in posterior enterocytes to increase citrate production/secretion, which promotes food intake and supports sperm maturation [49]. Thus, gut physiology is tuned by local sexual identity and inter-organ communication, actively coordinating sex-specific metabolic strategies.

Sex differences in gut physiology thus reflect evolved strategies tuned to reproductive investment. By modulating renewal, metabolism, and inter-organ communication, the ENS coordinates these strategies. We next focus on how ageing reshapes these sex-specific programmes and intersects with reproductive trajectories.

Sex-specific ageing trajectories of the *Enterarchon*

Sexual dimorphism and ageing highlight how *Enterarchon* functions shift with life history and physiological state. Ageing is often depicted as the progressive decline of physiological processes culminating in organ failure and death. In *Drosophila*, loss of intestinal barrier integrity—the so-called 'Smurf' phenotype [50]—provides a powerful, quantifiable proxy for systemic ageing.

Indeed, the ageing gut conforms to a biphasic model (Figure 4): an initial compensatory phase, during which organ function is maintained despite accumulating cellular damage, followed by a decompensation phase marked by systemic failure [50,51]. The Smurf transition, detected through increased intestinal permeability to a non-absorbable blue dye, predicts impending death across species, from days for nematodes and flies to weeks in mice and zebrafish [50−53]. This trajectory is sexually dimorphic, wherein the time dependence of Smurf incidence is stronger in females than males in *Drosophila*, an observation with no identified mechanism yet [50,52−54].

During the non-Smurf phase, homeostasis is sustained by elevated ISC activity and stress-responsive pathways [45,46]. Females rely on constitutively higher ISC proliferation to maintain turnover and organ size [40], whereas males depend more on stress-induced regenerative programmes [54]. Early features include sporadic dysplasia, subtle microbial shifts, and increased inflammation [53,55−57]. The transition is asynchronous across sexes: females show widespread dysplasia, disrupted regionalisation, and increased systemic inflammation, while males exhibit more gradual epithelial decline [54,58,59].

Sustained ISC proliferation, required for post-mating remodelling, renders females more susceptible to age-associated hyperplasia: with age, their hyperproliferative midgut becomes dysregulated, epithelial architecture and regional identity are lost, and barrier function fails [54,56,58]. These hallmarks are more pronounced in females, consistent with their higher baseline ISC activity, although direct evidence of proliferative exhaustion is lacking. In mammals, colorectal cancer incidence is higher in males, but females more often develop right-sided cancer [60], suggesting that sex-specific stem-cell dynamics and ageing differentially shape regional risk. By contrast, males—whose ISCs divide less frequently—maintain gut integrity longer yet cope less well with stress; they are more infection-prone [54], suggesting a trade-off between regenerative plasticity and resilience.

Life-history factors further shape this trajectory: age and sex influence ENS cellular composition and transcriptional states [61], and rodent work indicates that enteric neurons are selectively vulnerable to age-related degeneration [62], which can be partially prevented by caloric restriction. In neurodegenerative disorders such as Parkinson's disease, α-synuclein-containing Lewy body aggregates have been described in enteric plexuses, reinforcing the ENS as a key organ of clinically relevant ageing-associated pathologies [63].

This sexually dimorphic landscape reveals opportunities for targeted interventions. When initiated early, dietary restriction (DR) delays intestinal decline by reducing ISC hyperactivity in females, whereas males already maintain gut integrity with age and show little additional benefit [54,64]. DR changes energy metabolism and improves barrier integrity and delays Smurf onset [50,65,66]. Consistently, glycogen and triglycerides decrease with age in female flies, with an abrupt drop in Smurfs [50,66,67], showing a clear metabolic decline, depletion of energy stores, and loss of reproductive competence upon ageing (Figure 4). However, efficacy in delaying Smurf onset once decompensation begins remains unclear, highlighting the need for early biomarkers of gut ageing.

Thus, intestinal ageing is sex-specific in its modalities; understanding these strategies provides a framework for sex-specific interventions targeting gut ageing and systemic health.

Outlook

We reframed the intestine as an *Enterarchon*—an ancient epithelial–neuronal interface—and the ENS as a *visceral* brain-like regulator of internal state. Gut-embedded circuits arose on an early epithelial/peptidergic scaffold, diversified, and converged on peristalsis, interoception, and defence. In flies and mammals, enteric subtypes couple internal state to gut function [68], integrate endocrine–immune signals, and remodel with sex and age. Comparative single-cell atlases, circuit mapping, state-resolved physiology (including sex/reproductive status), and early biomarkers of barrier decline should test this framework and identify targets to extend gut function and healthspan.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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